

IDENTIFYING PATIENTS’ LEVELS OF CONSCIOUSNESS UNDER ANESTHESIA

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ABSTRACT. In this paper we present a two-stage machine learning pipeline for the estimation and prediction of patients’ pharmacological parameters relating to the induction anesthetic propofol. The first stage is to individually model the physical response that one thousand patients had to anesthesia during their surgical procedures. The second stage is to identify correlations between the biological demographic of a patient and how they responded to the anesthetic.

1. MOTIVATION AND PROBLEM STATEMENT

Since their advent, intravenous anesthetic induction agents have become an essential pillar supporting modern clinical medicine. Propofol is currently the most commonly used general anesthetic. Its ability to safely and rapidly transition patients to a state of unconsciousness and maintain that state throughout surgery enables over 300 million surgical procedures globally each year which would otherwise be prohibitively painful or life-threatening [MLH⁺15]. Though the exact norms for propofol administration vary around the world, broadly speaking dosages are determined by some biological criteria (weight in the U.S.) and corresponding population averages. Uniformly, however, automated infusion methods are separated from bispectral index (BIS) monitoring systems, which register a patient’s estimated state of consciousness throughout surgery [DA13]. This disconnect is due to the unreliability and noise in BIS readings, making them a poor dosage controller. A complete reliance on population averages increases the risk of incorrect dosages, which may have severe clinical consequences. Excessive propofol exposes patients to toxic effects, while under-dosing risks intraoperative awareness with explicit recall. This latter phenomenon is reported in 1-2 per 1,000 cases of general anesthesia, and may result in lasting psychological trauma [FMSP26, PAB⁺14]. Truly personalized dosage policies require an understanding of a patient’s individual pharmacokinetic parameters (how well their body metabolizes propofol) as well as their pharmacodynamic parameters (how sensitive their body is to propofol), in conjunction with continuous neurological state readings. This personalization is mediated manually by experienced anesthesiologists; no reliable automatic control system exists to personalize propofol dosage.

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Our objective is first, to extract personalized pharmacological parameters for a sample of patients, and second, to predict these profiles using preoperative demographic information. We aim to accomplish these objectives by unsupervised and supervised machine learning techniques, respectively. In so doing, we present a tool for a more accurate live prediction of a patient’s true neurological state during surgery, an important component in a personalized dosage control system.

2. DATA

2.1. Dataset Overview. We utilized the VitalDB public dataset, developed by Dr. Hyung-Chul Lee and Dr. Chul-Woo Jung at Seoul National University Hospital. This dataset was designed specifically to train machine learning models on perioperative clinical data, and is highly reputable [LPY⁺22, LJ18]. Though it is limited only to patients of the Seoul National University Hospital, its size (including 6,388 surgical operations) and scope (up to 184 concurrent continuous numerical vital signs per operation), together with its accessible Python-based API make it exceptionally useful for answering our research questions. We could identify no other dataset of its size that provided the necessary BIS readings, continuous propofol administration rates, and comprehensive preoperative patient variables. Its combination of static demographics data and dynamic time-series data lend this dataset exceptionally well to our two-phase machine learning approach to accomplishing our objective.

The raw VitalDB data is composed of two distinct categories: intraoperative time-series tracks with second-by-second resolution (for parameter extraction) and preoperative static data (for demographic regression). Table 1 summarizes the primary features utilized in this project.

Category	Feature	Description & Clinical Relevance
Intraoperative	BIS	EEG-derived measure of consciousness (0-100 scale); observation space (z_k)
	Propofol_Rate	Propofol infusion rate ($\mu\text{g/kg/min}$); control input (u_k)
Preoperative	Demographics	Age, Sex, Height, Weight (standard units)

TABLE 1. Summary of Selected VitalDB Features [LJ18]

2.2. Data Preprocessing. Cases were filtered to general surgeries containing the desired time-series and static data. Cases that used a medley

of induction anesthetics were discarded. For computational feasibility, a random subset of 1000 of the remaining cases was selected to proceed.

The raw BIS data required cleaning primarily on account of 0 readings at the endpoints of each surgery (before and after sensors were connected) and at intermediate values (when the sensor is temporarily displaced). The endpoints were cut from our data, as this appeared to represent a period of time before each operation had begun in our dataset. Rather than relying on standard interpolation to address the intermediate 0 entries, we substituted these with NaN values. Our unsupervised learning method utilizes the Kalman filter update step to address missing observations.

Also in the interest of computational feasibility, we widened the time resolution of our intraoperative data by a factor of 10, resulting in 10-second discrete time steps ($\Delta t = 10$). This involved careful verification that no important propofol infusion data was lost. As a final preprocessing step, the BIS index was shifted by -95, anchoring the fully awake state at zero. This was important for the dynamics of our state-space model used in parameter extraction.

Phase 2 required additional cleaning in order to train a random forest. The forest was trained to use the features of age, sex, height, and weight of the patients as provided by VitalDB in order to predict the parameters that patient would have in the previous model. The features only needed minimal cleaning: some attributes were converted from a string into an integer format. However, the classification values were more difficult: random forests give discrete outputs, so we had to take the continuous range that the parameters for the state transition scalar and the impact of propofol could be and first organize them into bins based on what values they held. Further, rather than directly use the values of the state transition parameters, we chose to take the inverse of the log of the values to decrease the risk of imprecision due to floating point error.

2.3. Data Splitting. Our Phase 1 methodology required no train/test split, as this involved an unsupervised learning method.

For the supervised machine learning component in Phase 2, after discarding extreme outliers we partitioned our data into an 80% training set and a 20% testing set. We randomized which data points were in which set to reduce potential issues.

3. METHODS

3.1. Phase 1: Pharmacological Parameter Extraction. To extract individualized pharmacokinetic and pharmacodynamic parameters from VitalDB’s time-series surgical data, we represented the evolution of a patient’s state of consciousness at any given time during surgery (x_t) with a one-dimensional stochastic discrete-time linear dynamical system. This state-space representation is ideal for its physical interpretability and computational efficiency with VitalDB’s time-series data. The state is continuously

driven by the known control of propofol infusion rate (u_t) and it is indirectly observed by means of the BIS monitor readouts (z_t). Altogether, this state-space model is given:

$$(1) \quad x_t = Fx_{t-1} + Gu_t + q_t$$

$$(2) \quad z_t = Hx_t + r_t$$

F represents the state transition scalar. If the propofol pump is disabled, we would expect that a patient’s consciousness level would increase asymptotically towards a state of restful awareness. We note again our BIS shift in data preprocessing 2.2, which was necessary to acquire this convergence behavior. Although biologically we would expect F to represent a composition of multiple factors, for the purposes of this project we treat F as a patient’s pharmacokinetic parameter representing propofol metabolism. Hence, we may assume that $F \in (0, 1)$. G is taken to represent the effect of the control input on the state, or a patient’s pharmacodynamic parameter dictating their sensitivity to the propofol. Because the anesthetic will drive a patient towards unconsciousness, we expect that $G < 0$. $H = 1$ maps the hidden biological state to the BIS scale of consciousness level. The terms q_t and r_t represent biological noise (random fluctuations within the brain) and measurement noise respectively, and are characterized by variances Q and R .

To accomplish our phase 1 objective of extracting each patient’s pharmacological parameters, we needed to fit the parameters F , G , Q , and R to our data. Initial attempts failed due to incorrect assumptions about our model and misunderstandings of our mathematical tools, however they proved to be instructive. See 3.1.1 for more details. Ultimately, our findings in these initial attempts led us to fix $Q = 0.05$ and $R = 25$ globally for all cases based on empirical data [ASK02]. In addition to empirical support for these choices, we found them to reasonably match our expectations of smooth neurological state transitions and observed noise in BIS readings. It should be noted however that this failure to extract case-specific noise parameters is a great weakness of our model. Our results will consequently stem from a baseline noise expectation rather than the true noise of each given operation, and hence the extracted pharmacological parameters are best understood in terms of relative value, rather than absolute. We then iteratively used the Kalman Filter to estimate the model’s hidden states and maximized the corresponding log-likelihood to fit F and G to each patient. It is noteworthy that gradient descent optimization methods like BFGS failed in this setting. We believe this was due to local minima. Instead, we employed a minimization method called differential evolution. This is a non-gradient stochastic method which operates by iteratively evaluating an objective function over a large area of candidate points [Ref]. Although this method is more computationally expensive, it served better to find global minimizers for F and

G for each patient. We met the cost of this optimization with a parallelized training scheme over our sample. Finally, we extracted pharmacological parameter estimates for each patient.

3.1.1. *Failed Attempts.*

(i). Kalman Filter. In our initial naive attempts, our Kalman Filter state estimates drastically overfit our BIS readings. Upon reflection, we realized that noise parameters maximizing the log-likelihood of all parameters case by case would assume high biological noise and that our system dynamics would depend almost entirely on our observations. We further reasoned that measurement noise must be fairly consisted from case to case, as BIS monitoring systems are standardized, and that biological noise should have negligible variability in sedated patients. From here, we endeavored to optimize our noise parameters across the entire sample and our pharmacological parameters patient by patient in an alternating optimization structure. However, even in this formulation our parameters manifested the aforementioned behavior, maximizing biological noise and minimizing measurement noise. Further evaluation led us to understand a key assumption of our mathematics that the surgical process violated: our measurement noise is highly non-gaussian. This should have been expected, as sudden bumps or impulses from electrocautery devices are prone to result in many instances of abrupt, high-magnitude noise in a BIS monitor [ASK02]. By fixing Q and R within empirically supported bounds we were able to regularize state estimation, significantly damping the effect of non-gaussian surgical noise and coaxing out the desired pharmacological parameters.

(ii). Hidden Markov Model. Additionally, we attempted to create a Hidden Markov Model to try and predict the hidden states of patient awareness during their surgeries. There was moderate success in a 1-dimensional model that took in only BIS data and was able to make fairly accurate predictions after training based on clinical thresholds. However when attempting to push the model further by adding in the rate of propofol injected, it became unstable and inaccurate as seen in Figure 1. Due to these limitations, we concluded that while a HMM could be useful for predicting hidden states, it did not lend itself to effectively providing the information necessary for our greater goals.

3.2. Phase 2: Demographic Predictive Modeling. In Phase 2, we searched for a correlation between patients' demographic information and their respective pharmacological parameter labels found in phase 1. We trained a random forest to accomplish this because forests can efficiently sort on multiple features.

The random forest was trained to use demographic features to predict the parameters such a patient would have. The features used were the age, sex, height, and weight of the patient. The forest predicted what parameter values the patient would have for the Phase 1 model using these features. Since forests give discrete outputs, this forest predicts one of several possible

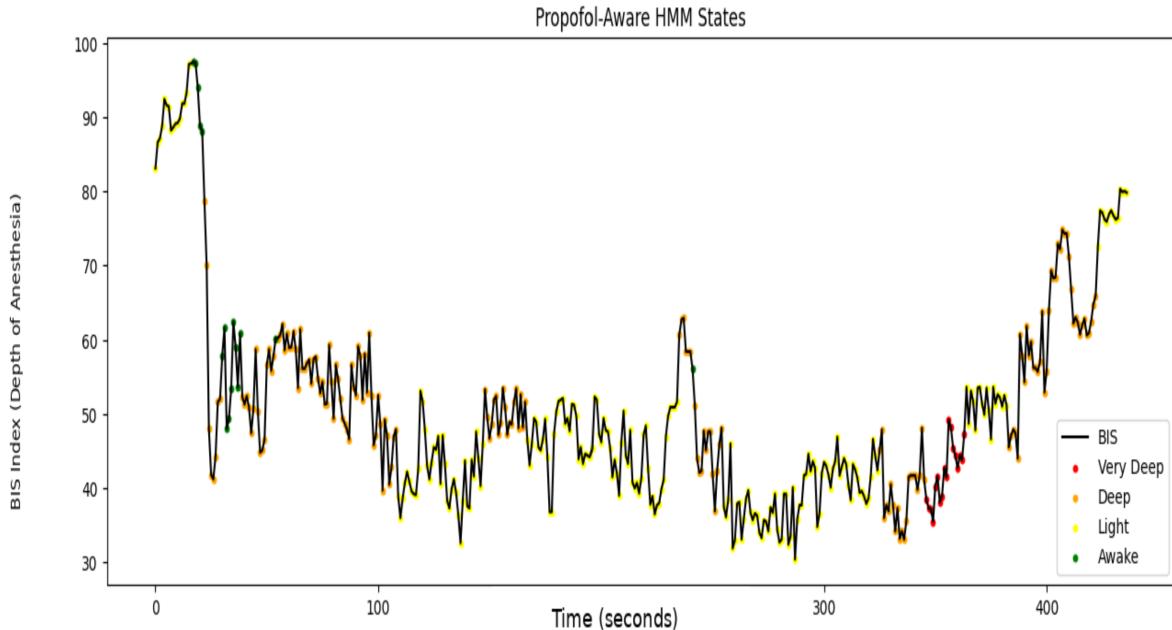


FIGURE 1. Hidden Markov Model that takes in both BIS levels and Propofol injection rate. Consistently underrated and overrated patient’s awareness level

values for both the state transition scalar and the impact of propofol scalar parameters.

The result is a random forest that can provide a similar prediction as the model in Phase 1 can train to find, but without requiring additional training or data from a surgery involving the patient. Instead, Phase 2 only requires someone’s age, sex, height, and weight to predict the individual’s pharmacokinetic and pharmacodynamic parameters. Since gathering demographic information is much simpler, more common, and less risky than directly gathering data during a surgery, this allows similar predictive power as the model in Phase 1 provides, but with general applicability on virtually any patient.

4. RESULTS AND ANALYSIS

4.1. Phase 1: Pharmacological Parameter Extraction. Our approach proved successful in estimating patients’ personal pharmacological parameters as well as their intraoperative neurological state. It is the ability of the Kalman Filter to estimate this state in real time, given their personal

pharmacological parameters, that would make this project a valuable tool for designating personalized propofol dosage policies. In the interest of clinically interpretable validation metrics for our method, we determined to use the Mean Absolute Error (MAE) of the Kalman Filter's estimated states against the raw BIS index time-series data. We would hope for this to be nonzero, as raw BIS readings are noisy and at times inaccurate. We would also hope that the average differences between the BIS readings and our state estimates would be tightly distributed, as this promotes our model's generalizability. Figure 2 shows our sample's MAE distribution has these desired qualities. Figure 2 displays the our state estimation with a 95% confidence interval against BIS readings for caseid 5332. This example characterizes our results for state estimations of other patients.

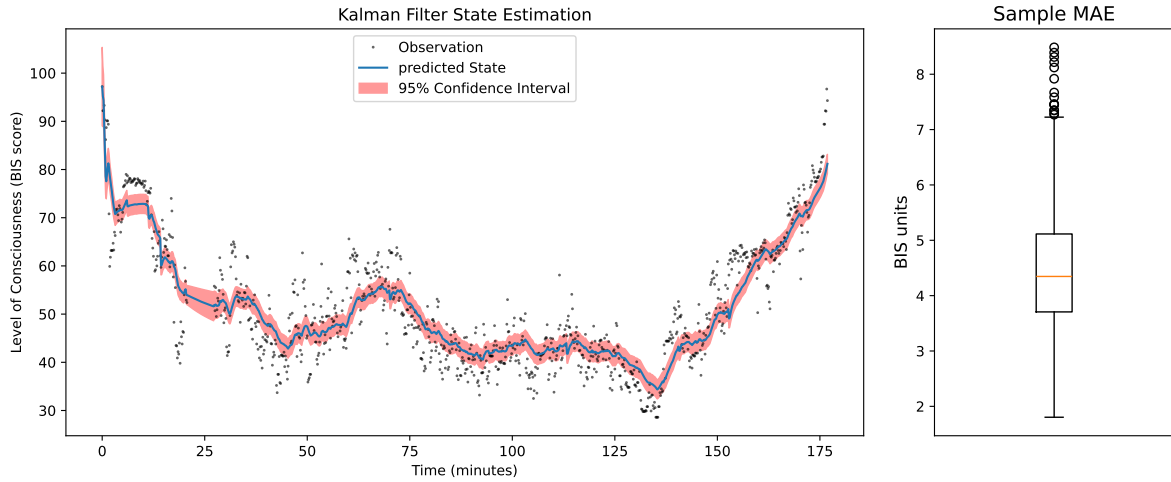


FIGURE 2. **Left:** Intraoperative BIS trajectory tracking for a representative patient. The plot displays raw clinical observations (black dots), the smoothed Kalman filter state estimate (solid blue line), and the associated 95% confidence interval (red shading). The mean width of this confidence interval (≈ 4.08 BIS points) demonstrates a significant reduction in variance compared to raw sensor noise. **Right:** Box plot illustrating the distribution of the Mean Absolute Error (MAE) across our patient sample

The F and G parameters extracted at this stage displayed fairly tidy distributions. Of our initial sample of 1000 patients, three yielded biologically impossible parameters (zero metabolism of propofol, spontaneous propofol production, and propofol resulting in an increased state of consciousness). We further eliminated eleven outliers. We defined our outliers using the IQR method with a 2.5 IQR scalar. We wanted to maintain high risk patients, as identifying these would be a crowning achievement of phase 2, but we

also wished to eliminate parameters resulting from flawed data. Visual inspection suggests that the outliers we filtered out resulted at least partially from abnormalities in the data. However, some of these outliers could have resulted from an anesthesiologist’s direct administration of a bolus of anesthetics during surgery, which is not represented in the dataset. This is an additional weakness of our model. Altogether, the 14 cases lost in this stage of cleaning seemed to be acceptable losses for our data requirements in phase 2. After this stage of data cleaning, we further processed our parameters to align with more standardized measurements. This involved converting the F parameter to a propofol half-life measurement, which is standard for expressing metabolic parameters [SSA18]. This follows from a straightforward formula $t_{\text{half}} = \frac{\ln(\frac{1}{2})}{\ln(F)}$. We also negated the G parameter to represent this as sensitivity in positive terms. Summary statistics may be seen at Table 2, and a density plot may be seen in Figure 3.

Statistic	Half-Life (minutes)	Sensitivity (BIS units per $\mu\text{g/kg/min}$)
Count	986	986
Mean	25.21	0.0147
Std. Deviation	23.28	0.0051
Minimum	8.99	0.0012
25th Percentile	16.57	0.0111
Median (50%)	21.01	0.0140
75th Percentile	26.64	0.0178
Maximum	310.93	0.0329

TABLE 2. **Summary Statistics of Extracted Pharmacokinetic Parameters.** Data represents the final analytical cohort ($N = 986$) following the removal of biologically impossible results and extreme statistical outliers.

4.2. Phase 2: Demographic Predictive Modeling. We used 80 percent of the patients’ data to train the forest, so the remaining 20 percent could be used to test its performance. We used a 1-norm comparing the actual category of each parameter with the assignment it received from the forest’s analysis. The result was that on average, each prediction was within 0.8275 of the correct bin. Since each patient received two parameters, this means any given patient is likely classified within one bin of the most correct option. Since each bin covers a relatively tight range, this means using the corresponding parameters of the prediction results in a decent model for that patient.

This model does have limitations. The most obvious being that the random forest predictions are on a bin rather than an exact value. Thus, the

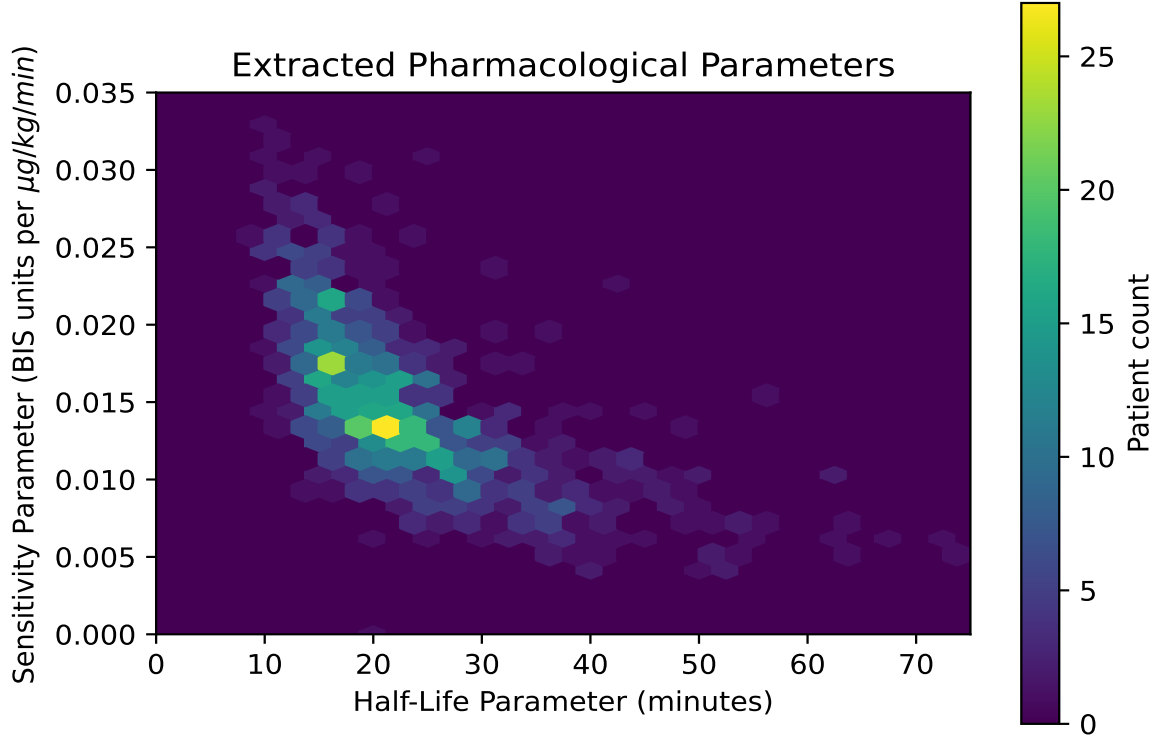


FIGURE 3. Hexbin density plot illustrating the distribution of the extracted pharmacological parameters (derived from F and G) across the 1,000-patient cohort. Parameters were optimized via Maximum Likelihood Estimation and Differential Evolution utilizing a Kalman Filter.

F and G parameters that the random forest predicts have a greater degree of error than what is generated in Phase 1, even without accounting for the probability of being classified to a suboptimal bin. To be precise, the relative error of the bins for the F parameters is roughly 0.000413%, and for the G parameters is roughly 0.0852%. Thus, the problem that results from being even 5 bins off in both parameters is still less than a 1% relative error from correct. But this error is always present, and could be compounded with each time step when used in the model from Phase 1.

5. ETHICAL CONSIDERATIONS

The use of the Vital Database itself is ethically clean: personal information has been made anonymous and the entire dataset has been publicly released to the world. Thus, there are no clear problems with the data we are analyzing, nor is there a clear danger of feedback loops.

Furthermore, since our analysis is only practical in the context of a surgical procedure, there is low risk of these models being used by anyone other than trained medical professionals. Thus, it is reasonable to assume that any doctor who is already trusted to perform a surgery should be given as much valuable information as possible to perform optimally. This does raise one potential risk: if the model fails in a situation the doctor(s) trusted it, this could result in significant harm to the patient. This danger can be mitigated by rigorous and independent testing of this model before adopting use. In any case, it is our feeling that there is a moral responsibility to be borne in any context relating to choices that affect the health and safety of human life. This cannot be set upon a machine or an algorithm. Were this machine learning pipeline to be utilized in a clinical setting, the moral responsibility for its effects would be shared between ourselves as its designers as well as the medical professionals who implement it.

6. CONCLUSIONS

Altogether, our results show genuine proof of concept in meeting our target objectives. Our extraction of pharmacological parameters from time-series data, and corresponding Kalman Filter state estimation, show improvement over raw monitors of neurological state. Furthermore, our regressor demonstrates promise as it relates to predicting those parameters *a priori*. However, our approach does have several prominent limitations, most notably in its global assumptions of static noise constants, its omission of potential anesthesiologist intervention through direct bolus, and its scope including only propofol. Obvious directions for improvement include expanding the model to account for the effects of additional anesthetics and expanding the dimensionality of our observation and state variables to utilize additional time-series data available through VitalDB.

Especially important to note is that all the work done here is still purely theoretical—rigorous real-world testing is important to verify these findings as accurate before they can be used in practical applications. This is something beyond the scope of the authors’ ability to do.

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